CANADIAN NEMO OCEAN MODELLING FORUM
COMMUNITY OF PRACTICE

Canadian Ocean Modelling workshop (11-13 May 2026) summary

The Canadian NEMO Ocean Modelling Community of Practice held its second workshop from the 11th to the 13th of May, 2026, online. There were a total of 56 participants from the CoP community. The first and third days consisted of two 2-hour discussion sessions on topics that were voted on as most relevant and of interest by the community. The topics were: 1) Ocean modelling development: Globally and in Canada, 2) AI Integration into NEMO modelling and analysis processes, 3) Biogeochemical models development and directions, and 4) Cross-institute coordination of NEMO development. The second day consisted of an Early Career Researcher workshop on the use of Artificial Intelligence and Machine Learning in Earth Sciences.

Transcripts of the discussions, videos of day 2 and 3 and meeting minutes can be found: https://canadian-nemo-ocean-modelling-forum-commuity-of-practice.readthedocs.io/en/latest/Workshop2026.html

1. Ocean modelling development: Globally and in Canada- discussion leaders Paul G. Myers and Nicolas Lambert

Paul Myers and Nicolas Lambert led discussions on spatial and temporal scales of ocean modelling, with participants sharing challenges around forcing data, model configurations, and diagnostic tools. The group explored ways to better coordinate and share code, forcing files, and analysis tools across different Canadian institutions using NEMO. Key topics included the challenges of transitioning to NEMO version 5, the need for consistent model configurations and forcing products, and the development of diagnostic tools to analyse model output. Participants also discussed the integration of adjacent models, such as CICE, SI^3 and biogeochemical components, as well as the use of couplers and I/O servers. The session concluded with agreement on the need for better coordination of tools and resources, particularly through GitHub organisations and shared repositories. The group agreed that while code consolidation would be beneficial, it requires significant resources and coordination, with some suggesting alternative approaches like making all code bases public-facing to enable organic sharing and collaboration.

Paul highlighted the need for extended forcing products to run hindcast models beyond 2019-2022, particularly for river runoff and Greenland melt data. Nicolas then presented on diagnostic tools, showcasing existing NEMO-based tools including CDFTOOLS, a Python library, and Stephanie Taylor's NEMO SCOPE code, while raising questions about reproducibility, portability, and community coordination of these tools. Suggestions of creating a central GitHub repository to share diagnostic code and configurations across the community.

2. AI Integration into NEMO modelling and analysis processes- discussion leaders Amber Holdsworth and Andrew Shao

The second session of the first day focused on AI integration into NEMO modelling and analysis processes. Andrew Shao presented an overview of AI and machine learning applications in ocean-atmospheric fluid modelling, highlighting the rise of GPU computing and its impact on AI training and inference. He discussed the advantages of AI surrogate models, including faster computation times and reduced development time compared to traditional numerical models. The group also explored the potential for using AI models to bypass traditional data assimilation steps in weather forecasting. Andrew emphasised the importance of including ocean representations in AI surrogate models for improved stability and long-term performance. The discussion concluded with considerations about when and where to use AI emulators versus numerical simulations, and the potential benefits of hybrid methodologies.

Andrew presented on incorporating AI and machine learning into ocean modelling simulations, discussing how AI can complement traditional numerical models rather than replace them. The discussion touched on the benefits of using AI for hypothesis testing and emulating complex processes that are difficult to model from first principles, with Fred Dupont providing examples of using emulators for processes like light penetration and wave action. Amber Holdsworth raised a question about whether machine learning approaches might eventually incorporate numerical components, leading to a discussion about the evolution toward hybrid models that combine traditional numerics with machine learning techniques.

3. ECR workshop: AI/ML introduction - Andrew Shao and Kirsten Mayer

Day 2 was an Early Career Researcher workshop about an Introduction to AI and machine learning for the Canadian Ocean Modelling, led by Kirsten Mayer from NCAR and Andrew Shao from HPE. Kirsten Mayer provided a comprehensive overview of artificial neural networks, covering fundamental concepts including machine learning terminology, supervised learning approaches, and the architecture of neural networks. The presentation included detailed explanations of key components like weights, biases, activation functions, and gradient descent optimisation. After the lecture portion, participants worked on an exercise using Google Colab to train their own neural networks to predict El Niño or La Niña conditions from tropical sea surface temperature data. The session included extensive Q&A discussions about hyperparameters, model architecture, overfitting, and practical implementation challenges, with participants sharing their experiences and questions about the hands-on exercise.

4. Biogeochemical models development and directions- discussion leaders Elise Olson and Inge Deschepper

The first session of the third day reviewed recent developments in biogeochemical (BGC) models across Canadian groups and discussed national BGC priorities and modelling needs. Groups from across Canada presented a quick summary of their current work and BGC model development.

Participants agreed national BGC modelling priorities include ecosystem climate impacts, northern carbon-cycle representation, nature-based carbon removal, coastal forecasting, vulnerability assessment, and compound stressor projections. Development needs emphasised constraining rate responses (remineralisation, production, grazing, mortality) to temperature and acidification; attendees judged many rate sensitivities currently under-constrained.

The group identified persistent gaps between observationalists and modellers driven by mismatched requests, limited mechanistic coverage in field data, and limited personnel bandwidth for iterative engagement. Participants proposed structured working groups alternating modeller-only and observation-inclusive meetings, routine metadata paragraphs in observational collations explaining modelling relevance, and targeted workshops to align measurements with modelling needs. Mesocosm and lab experiments were recognised as useful sources of process data, while systemic funding and structural barriers were flagged as hindrances to cross-institution collaboration and student placements. ML/AI was acknowledged as promising for extrapolating sparse inputs and for reanalysis work, but limitations from sparse training data remain.

5. Cross-institute coordination of NEMO development- discussion leaders Natasha Ridenour and Heather Andres

Cross-institute coordination focused on shared technical artefacts, capacity shortfalls, and governance of shared resources. Participants catalogued common technical pain points (forcing standardisation, high-resolution atmospheric forcing mismatches, river routing, NEMO5 migration) and widespread hiring and retention problems for skilled modellers. The group favoured sharing small, high-value artefacts (diagnostics, namelists, parameter-testing cases, forcing snippets) and curated observational products with QA/QC to ease model evaluation. A neutral shared Git organisation was proposed and gained growing agreement as a practical starting point for hosting code, tools, discussion pages, and short problem→solution write-ups; public NEMO forums were recommended for NEMO-specific questions to leverage international resources while maintaining a Canada-focused space for domestic topics and early-career visibility.

Decisions and commitments recorded included moving forward with a shared repository organisation, encouraging participants to populate the Community of Practice site with institutional descriptions and model setups, and volunteers to contribute initial content such as parameter tests and example modifications. Persistent unresolved constraints remain: limited personnel and funding, platform hosting and long-term maintenance choices, data-sharing legal/formatting barriers, and technical issues migrating to NEMO5 and producing ensemble/high-resolution products.